Journal of Membrane Science 738 (2026) 124851

Contents lists available at ScienceDirect

Journal of Membrane Science

journal homepage: www.elsevier.com/locate/memsci

Hydrophobization of carboxymethyl cellulose by Passerini reaction: towards films with improved water vapor barrier properties

Clémence Vuillet [a], Valérie Guillard [b], Hélène Angellier-Coussy [b], Guillaume Sudre [a], Fabrice Gouanvé [a,*], Etienne Fleury [a], Aurélia Charlot [a,**]

[a] Université Claude Bernard Lyon 1, INSA Lyon, Université Jean Monnet, CNRS UMR 5223, Ingénierie des Matériaux Polymères, F-69621 Cédex, France
[b] JRU IATE, University of Montpellier, INRAE, Institut Agro, 2 Place Pierre Viala, 34060, Montpellier, France

ARTICLE INFO

Keywords:
Carboxymethyl cellulose
Passerini
Barrier
Dynamic vapor sorption
Hydrophobization

ABSTRACT

To address the need for developing sustainable materials with effective barriers to water vapor, this work explores the potential of self-supported films made of carboxymethyl cellulose (CMC) functionalized by the Passerini three-component reaction. Aliphatic aldehydes and *tert*-butyl isocyanide were grafted onto CMC backbone to yield dually hydrophobized derivatives. These modified polysaccharides were processed into films by solvent casting and their water vapor transfer properties were examined. Small-angle X-ray scattering (SAXS) experiments revealed a nanoscale organization for Passerini-modified films, attributed to the formation of hydrophobic domains characterized by a nanometric interdomain spacing. Dynamic vapor sorption (DVS) analysis demonstrated a significant reduction in moisture content water for modified films, from $\simeq 1.1$ g g^{-1} for unmodified CMC to below 0.5 g g^{-1} for hydrophobized derivatives at $a_w = 0.95$. Guggenheim-Anderson-deBoer (GAB) and Zimm-Lundberg modeling showed a decrease in sorption site availability from 3 to $\simeq 1.5$ per glucose unit, while the size of water clusters was significantly increased. The Passerini modification also resulted in a substantial decrease of water vapor permeability (WVP) from 44 000 barrer to 2500 barrer at $a_w = 0.5$. These results unequivocally underpin the benefits of the Passerini functionalization which allows to enhance the water vapor barrier performance. The findings highlight the potential of such a reaction for developing next-generation and bio-based packaging materials with tailored water vapor barrier properties.

1. Introduction

The packaging industry is currently undergoing a major transformation, driven by the need to address growing environmental concerns through the development of bio-based and biodegradable materials able to substitute conventional plastics [1]. Particularly, polysaccharides have garnered significant attention, as they offer attractive properties of biodegradability, renewability and wide-spread availability [2]. In the literature, polysaccharides extracted from plants (starch [3], pectin [4]), animals (chitosan [5]), algae (alginate [6], carrageenan [7]), or bacteria (gellan gum [8]) have been processed into self-standing films. Among polysaccharide derivatives, carboxymethylcellulose (CMC) is a carboxymethylated anionic ether derivative of cellulose, the most abundant organic resource on earth. CMC stands out for its excellent film-forming properties, and remarkable oxygen and lipid barrier properties [9]. However, generally, due to their inherent hydrophilic character, polysaccharide-based films (PBFs) show poor stability in the presence of water or water vapor, which significantly restricts their use in applications [10,11]. Moreover, their performance is generally altered at higher temperature and high relative humidity, which is problematic for the storage of packaging in tropical conditions, particularly in humid geographic zones. Indeed, the strong affinity of polysaccharides for water leads to increase permeation of water molecules through the film.

To address this limitation, various strategies have been investigated [12,13]. Among physical approaches, the incorporation of fillers or additives has proven effective to reduce water sensitivity of neat polysaccharides. In particular, nanocomposites have drawn significant interest, with many studies focusing on the addition of fillers that act as impermeable obstacles, allowing to decrease the water vapor permeability (WVP) through a tortuosity effect and to decrease the water vapor diffusion coefficient [14]. Clays minerals, such as montmorillonites

* Corresponding author.
** Corresponding author.
E-mail address: aurelia.charlot@insa-lyon.fr (A. Charlot).

https://doi.org/10.1016/j.memsci.2025.124851
Received 17 July 2025; Received in revised form 17 October 2025; Accepted 19 October 2025
Available online 28 October 2025

were already been used [15–17], but also bio-sourced organic fillers, such as cellulose nanocrystals (CNCs) and cellulose nanofibrils (CNFs) [18–21]. Indeed, the addition of such cellulose fillers enables the formation of a specific tortuous diffusion path created by the entanglements and the peculiar arrangement of the high aspect ratio of CNF, yielding a decrease of the diffusion of water molecules. Blending CMC with other molecular compounds was also shown to be promising. For instance, emulsions with oleic acid result in films with reduced WVP [22], while incorporation of hydrophobic cinnamaldehyde improves both water resistance and biocodal properties [23]. Furthermore, combining CMC with other polysaccharides such as starch [24], or pullulan [25] leads to mixed films with lowered WVP. These improvements can be attributed either to polymer-polymer interactions – such as hydrogen bonds between carboxyl and hydroxyl groups, which compete efficiently with the water-polymer affinity – or to the intrinsic barrier properties of the added polysaccharide itself. Chemical modification of polysaccharide is another valuable route to enhance barrier effects against water. Cross-linking of CMC, either through surface post-treatments (such as UV-induced cross-linking with sodium benzoate, or chemical crosslinking by exposing CMC/gelatin films to glutaraldehyde vapor) [26], or treatments in solution prior to the casting step, have been shown to significantly increase the water contact angles and decrease both water solubility, and WVP. These effects are attributed to a denser and a more ordered arrangement of polymer chains linked to an increase in crystallinity, as evidenced by the formation of diffraction peaks visible in X-ray patterns. The creation of crosslinking within the polymer network increases the tortuosity of the diffusion pathway of water molecules. Chemical crosslinking of CMC through esterification with citric acid in the presence of polymers such as polyvinyl alcohol (PVA) [27], or guar gum [28] has yielded films with enhanced stability against moisture. Moreover, pectin [29] and galactomannan [30] were esterified with acetic anhydride and were engineered as films that exhibit high water contact angles (>100°). Starch acetylation also allowed the formation of films with reduced water solubility, and lowered WVP [31]. These examples show that the chemical modification of polysaccharides, in particular the grafting of hydrophobic substituents is a valuable and efficient route to decrease the polarity of the films (and thus the sorption of water molecules) but also to create a specific tortuous diffusion path impeding the diffusion of small species, such as water molecules. However, to the best of our knowledge, chemical derivatization of CMC for end-uses as barrier films, has never been reported so far. To this end, the use of versatile, modular and sustainable chemical functionalization routes is highly desired to target polysaccharides derivatives with appropriated and tailorable (multi) functionalities.

In this frame, multicomponent reactions (MCRs) are powerful and combinatorial chemical tools that offer the possibility to concomitantly introduce multiple functionalities onto a polysaccharide backbone in a single step. Among MCRs, Passerini three-component reaction (P-3CR) is particularly well-suited to modify polysaccharides, since it involves the reaction between a carboxylic acid, a carbonyl (typically an aldehyde) and an isocyanide, yielding an α-acyloxycarboxamide ligation incorporating all the atoms of the raw precursors. Moreover, P-3CR is totally compatible with aqueous mild conditions, and is conducted in catalyst and coupling reagents-free conditions. This was first demonstrated by Crescenzi et al. in 1999, who generated cross-linked hydrogels by applying P-3CR to various polysaccharides [32]. P-3CR was further employed to modify succinylated cellulose [33], cellulose dialdehyde [34], tempo-oxidized cellulose nanofibrils [35,36], and recently to periodate oxidized cellulose nanofibrils [37]. Our research team devoted great efforts to the functionalization of polysaccharides by P-3CR reaction. We proved the feasibility of this approach by modifying CMC with various aldehydes and isocyanides [38]. Then, the derivatization route was applied to periodate-oxidized nanofibrils of cellulose, to impart them with functionalities prone to react by aza-Michael and Huisgen cycloaddition reaction [39]. We have extended the Passerini modification to the use of ketones (instead of aldehydes) to prepare a novel set of CMC-based products with the aim of generating chemically cross-linked hydrogels [40]. More recently, we have reported the synthesis by P-3CR of CMC-based comb copolymers carrying both hydrophobic moieties and thermosensitive segments [41] and we have described the design of innovative families of alginate and CMC derivatives dually hydrophobized by various aliphatic substituents [42]. Such resulting derivatives, deposited onto substrates, display tunable wettability properties, whose hydrophobicity can be modulated by the structure of the grafted molecules.

Leveraging all these findings, our aim is to investigate the effect of the Passerini chemical functionalization on the relationship between the structure and the water vapor transfer properties of CMC-based self-supported films. For that purpose, different hydrophobic aldehydes and isocyanides were used as reactants and self-supported films were further prepared by casting method. The films were first optically examined in liquid water, alongside a detailed analysis of their internal structure using Small-Angle X-ray Scattering (SAXS) under different temperatures and various relative humidities. Water vapor transfer properties were assessed by the water vapor sorption properties *via* gravimetric analysis at 20 °C and across a wide range of water activities (a_w from 0 to 0.95). Water vapor permeability measurements were also performed. The Passerini modifications are expected to significantly impact the transport of small molecules within the self-supported films by altering polarity and creating a peculiar nanoscale structural organization provided by the twofold chemical grafting.

2. Experimental section

2.1. Materials

CMC was provided by Sigma Aldrich. It is found under the form of sodium carboxylate salts, with a degree of substitution in carboxylate groups of 1.2 (i.e. 1.2 carboxylate groups per one sugar unit). The mass average molecular mass M_w of this polysaccharide was determined by Size Exclusion Chromatography and is 250 kg mol^{-1}, with a dispersity of $Đ$ of 2.5. Butyraldehyde, (But), heptaldehyde (Hept), 5-norbornene-2-carboxaldehyde (Norb), and *tert*-butyl isocyanide (*Tert*B) were provided by Sigma Aldrich. Hydrochloric acid (HCl), acetone and isopropanol (IPA) were provided by Carlo Erba.

2.2. Passerini functionalization of CMC

A representative procedure for sample 1 in Table 1 is detailed as follows, and is based on a published procedure [42]. CMC (1.5 g, 5.81 mmol of COONa, 1 eq. COONa) was dissolved in 100 mL of deionized water, resulting in a 1.5 % (w/v) solution. The mixture was continuously stirred at room temperature (25 °C) for 24 h to ensure complete dissolution. A partial acidification degree (DS_{COOH}) of 0.7, corresponding to 0.7 in average of –COOH groups per one sugar unit (4.04 mmol), was targeted in order to reach a sufficient number of COOH groups prone to react in Passerini reaction, while maintaining the water solubility of the

Table 1
P-3CR reactions applied to CMC with various aldehydes and isocyanides: experimental conditions and reaction characteristics.

Entry	Sample[a]	*Eq* R_1[b]	*Eq* R_2[b]	*DS* (±5 %)[c]	Mass yield (%)[d]
1	CMC-But-*Tert*B	0.41	0.36	0.13	89
2	CMC-But-*Tert*B	0.82	0.72	0.31	99
3	CMC-Norb-*Tert*B	0.82	0.72	0.35	68
4	CMC-Hept-*Tert*B	0.82	0.72	0.17	84

[a] Reaction conditions: 1.5w/v of CMC, 18-20 h, 25 °C
[b] Equivalent of reagents with respect to available COOH functions.
[c] *DS*: degree of substitution, number of grafted substituents per one sugar unit, calculated by ^{1}H NMR (see Experimental section),
[d] Mass yield.

polysaccharide derivative. By assuming a complete reaction of HCl with COONa, a solution of hydrochloric acid at 1 M was gradually added to the reaction solution (corresponding to 4.04 mmol and 4.04 mL). The addition of HCl leads to a decrease of pH from 7.0 down to 3.7.

While maintaining magnetic stirring, butyraldehyde (300 μL, 0.82 molar equivalent relative to –COOH groups) and *tert*-butyl isocyanide (336 μL, 0.72 mol. eq. relative to –COOH groups) were introduced dropwise into the reaction mixture. The reaction proceeded at 25 °C for 18 h. Subsequently, the mixture was precipitated by adding it to 1 L of cold IPA. The resulting solid was collected by filtration using a fritted glass filter and washed four times with 150 mL of acetone. The purified product was then re-dissolved in deionized water and subjected to overnight stirring to ensure thorough purification. Finally, the solution was freeze-dried for 72 h. The final products were stored under inert atmosphere in sealed vials. A detailed explanation of abbreviations of the synthesized samples is provided in Fig. 1. For instance, CMC-But-TertB$_{0.13}$ (Table 1, line 1) refers to CMC grafted with butyraldehyde and *tert*-butyl-isocyanide at a DS of 0.13.

2.3. Preparation of self-supported CMC-based films

Self-supported films were prepared by casting aqueous solutions of CMC derivatives. Solutions were prepared by adding 375 mg of Passerini-modified CMC in 15 mL of distilled water, to obtain solutions at 25 g L^{-1}. They were left to dissolve overnight, and subsequently left to rest for 24 h. For each film, 12 mL of the solution was poured into 5 cm diameter Teflon molds. Drying was then carried out at 70 °C for 5 h to obtain the final films with thicknesses ranging from 110 ± 12 μm to 124 ± 16 μm; the films were stored in a desiccator to prevent moisture up-take. Thinner films with thickness ranging from 40 μm to 65 μm were obtained by casting 6 mL of solution into 5 cm diameter Teflon moldand were specifically used for the permeability measurements.

2.4. Characterization of Passerini-modified CMC-based products

2.4.1. Infrared analysis (IR) spectroscopy

FTIR spectra were acquired with a Nicolet iS10 (ThermoFischer Scientific) through attenuated reflectance mode (ATR) on diamond crystal. A total of 32 scans were acquired from 650 to 4000 cm^{-1}.

2.4.2. Nuclear magnetic resonance (NMR) spectroscopy

NMR spectra were acquired using a Bruker Avance 400 spectrometer. Used parameters were the following, ^{1}H: 400 MHz, 512 scans, $D_1 = 10$ s; ^{13}C: 400 MHz, 12288 scans, $D_1 = 3$ s; 2D HSQC and HMBC: 400 MHz, 32 scans, $D_1 = 2$ s. Analysis were mostly conducted on 10 g L^{-1} solutions in D_2O. However, if the water-solubility of products was not satisfying, analyses were conducted in 50/50 v/v D_2O/organic solvent mixtures at 323 K. TSP-d4 was introduced as internal reference.

Degree of substitution, referred as DS, corresponds to the average number of grafted α-acyloxycarboxamide function per sugar unit of CMC. DS values were determined from the ^{1}H NMR spectra using Equation (1) and Equation (2).

$$m_{graft} = \frac{m_{TSP-d4}}{M_{TSP-d4}} \times \frac{H_{TSP-d4}}{I_{TSP-d4}} \times \frac{I_{graft}}{H_{graft}} \times M_{graft} \quad (1)$$

$$DS = \frac{m_{graft}/M_{graft}}{(m_{total} - m_{graft})/M_{sugar\ unit}} \quad (2)$$

Here $m_{TSP\text{-}d4}$ refers to the mass of the internal reference TSP-d4, m_{total} represents the total mass of the polymer dissolved in the NMR tube, and m_{graft} is the mass of the grafted moiety, calculated by using Eq. (5). I_{graft} and $I_{TSP\text{-}d4}$ correspond to the integrals of well-defined peaks assigned to the chemical groups of the grafted moieties and to the internal reference (TSP-d4), respectively. $H_{TSP\text{-}d4}$ and H_{graft} correspond to the number of protons associated with these integrals. M values refer to the molar masses of TSP-d4, of a single sugar unit, and of the covalently grafted residue, where ($M_{graft} = M_w(aldehyde) + M_w(isocyanide) + 1$). The DS value enables the estimation of the theoretical mass of the product (m_{theo}) which considers the molar mass changes resulting due to the Passerini functionalization. This value is calculated using Equations (3) and (4).

$$m_{theo} = \frac{m_{init} * M_{P3CR}}{M_{sugar\ unit}} \quad (3)$$

$$M_{P3CR} = M_{sugar\ unit} + DS \times M_{graft} \quad (4)$$

Here, m_{init} is the mass of the initially weighed polysaccharide CMC, and $M_{sugar\ unit}$ is the molar mass of the sugar unit.

Finally, the mass yield is determined by dividing the experimentally obtained product mass (m_{exp}) by the theoretical product mass (m_{theo}) as described in Equation (5).

$$mass\ yield = \frac{m_{exp}}{m_{theo}} \times 100 \quad (5)$$

2.5. Dimensional stability of films in water by optical visualization

The films were cut into square pieces of approximately 0.5 cm. The samples were then immersed in distilled water, at 25 °C. Immersion experiments were conducted during predefined times (1 min, 4 h, 5 days), and photographs of the films were taken by using a photo booth equipped with controlled lighting conditions to document visual changes.

2.6. Small-angle X-ray scattering (SAXS)

The small-angle X-ray scattering patterns of CMC films were measured on the Xeuss 3.0 SAXS/WAXS instrument operating with a copper source (λ = 1.542 Å) in standard configurations at three sample-to-detector distances (37 cm, 90 cm, 180 cm). Films of 109 ± 12 μm to

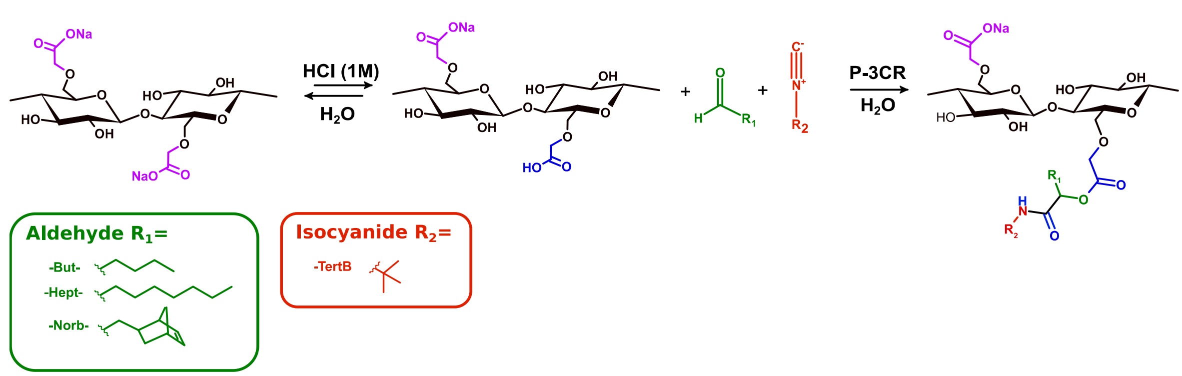

Fig. 1. Acidification of CMC and subsequent P-3CR with hydrophobic aldehydes (R_1) and hydrophobic isocyanide (R_2).

124 ± 16 μm thickness were cut into 5 × 10 mm^2 rectangles. A detector Dectris EIGER2 Si 1 M was used. Temperature and relative humidity were precisely controlled using the Xenocs humidity stage. Various water activities (a_w) were considered and a_w is defined as the ratio of the vapor pressure of water in equilibrium with the material (P), over the saturated vapor pressure of pure water (P_{sat}). a_w is numerically equivalent to the relative humidity (RH) divided by 100 ($a_w = RH/100$). Thus, SAXS measurements were conducted under four distinct successive environmental conditions: 21.3 °C/a_w = 0, 21.3 °C/a_w = 0.90, 45.5 °C/a_w = 0.90, 21.4 °C/a_w = 0.90. The objective was to observe if the increase of both temperature and water activities could induce a variation of the overall mobility and thus a modification of the internal structure. Two analyses in similar conditions (before and after the increase of T°) were undertaken to check the reversibility of the response.

Scattering data were normalized by transmitted intensity, and the signal of an empty cell was recorded and subtracted to each sample intensity. The I vs q patterns were obtained through normalization of the data by the exposure time and transmission and sample thickness, followed by azimuthal average.

2.7. Dynamic water vapor sorption (DVS)

2.7.1. Sorption isotherms

Water vapor sorption experiments were carried out at by using a dynamic vapor sorption (DVS) equipment (Surface Measurement System Ltd., London, U.K.) on self-supported films. The mass evolution at different a_w were monitored over time by using a Cahn microbalance. Each experiment began with a drying phase conducted in DVS equipment, during which the sample was exposed to dry air until reaching a constant mass. The sample was then subjected to increasing water activities (a_w) levels (0.1, 0.2, 0.3, 0.4, 0.5, 0.6, 0.7, 0.8, 0.9, and 0.95), at 20 °C. Water activity increments were applied once mass equilibrium was achieved, defined as mass variation kept below 0.0022 % per minute over a period of 10 min. Experiments were conducted in triplicate. Water vapor sorption isotherms were generated from the equilibrium moisture content (M) measured at each a_w level. This moisture content is the ratio of the difference between wet and dry mass to dry mass (M_w and M_d) (Equation (6)) and thus corresponds to the ratio between the mass of water sorbed and the mass of polymer, in $g_{water} \cdot g_{polymer}^{-1}$ (g.g^{-1} in the rest of the text).

$$M = \frac{M_w - M_d}{M_d} \tag{6}$$

The moisture content was then plotted as a function of water activity.

Based on the mass gain data, the average number of water molecules sorbed by a polymer unit, written Ni, is determined by Equation (7).

$$Ni = \frac{M \times M_{pol}}{M_{H2O}} \tag{7}$$

Where M_{pol} is the molar mass of the monomer unit, and M_{H2O} is the one of water.

2.7.2. Mathematical modeling

The mathematical description of sorption isotherms enables to get useful information on the sorption mode and interactions involved during sorption. Several mathematical models are proposed to describe the sorption isotherms. Among them, the Guggenheim-Anderson-de Boer (GAB) model, is considered to be the most appropriate model for water sorption in the case of cellulosic materials [43]. This model considers that water molecules condense layer by layer, and is described by Equation (8).

$$M = \frac{M_m.C_g.k.a_w}{(1 - k.a_w)(1 - k.a_w + C_g.k.a_w)} \tag{8}$$

Where M is the equilibrium water content (g.g^{-1} d b.), M_m the water content required to saturate all the primary sorption sites, defined here as the monolayer (g.g^{-1} d b.), C_g the Guggenheim constant related to the energy associated with the interactions between the penetrant molecules and the first sorption sites, k a corrective factor relative to the adsorption energies of the second and the subsequent layers.

The values of parameters leading to the best fit between the calculated curve and experimental curve were determined by using the software Table Curve 2D. Quality of fits were assessed through the Mean Relative Deviation (MRD) values, defined in Equation (9).

$$MRD(\%) = \frac{100}{N}\sum_{i=1}^{N}\frac{|m_i - m_{pi}|}{m_i} \tag{9}$$

In the case of hydrophilic materials, water concentration in the polymer at high water activities might induce relaxation of preexisting voids, resulting in swelling. The theory developed by Zimm and Lundberg is key to analyze water clustering. Their method is based on statistical mechanics, and provides an interpretation of the thermodynamic behavior of the polymer-solvent system in geometric isotherms. The resulting mathematical relation is represented in Equation (10).

$$\frac{G_S}{V_{water}} = -(1 - \Phi_{water})\left[\frac{\partial\left(\frac{a_w}{\Phi_{water}}\right)}{\partial a_w}\right]_{P,T} - 1 \tag{10}$$

Where G_S represents the cluster integral, and V_{water} and Φ_{water} respectively the partial molecular volume and the volume fraction of water within the polymer film. A value of G_S/V_{water} equal to −1 indicates a random distribution of water molecules. However, $G_S/V_{water} > -1$ indicates that the concentration of water in the neighborhood of a given water molecule is greater than the average concentration of water molecules in the whole polymer. In this case, the average size of water aggregates, referred to the mean cluster size (MCS), can be defined according to Equation (11).

$$MCS = 1 + \left[\frac{\Phi_{water}G_S}{V_{water}}\right] \tag{11}$$

The volume fraction of water within the polymer film is obtained from Equation (12).

$$\Phi_{water} = \left(1 + \frac{\rho_{water}}{M.\rho_{pol}}\right)^{-1} \tag{12}$$

Where ρ_{water} and ρ_{pol} are the densities of water and the polymer, respectively. The equation for calculating the MCS as a function of the GAB parameters is defined in Equation (13).

$$MCS = \frac{\left(\frac{\rho_{water}}{\rho_{pol}}\right)^2}{M^2\left(1 + \frac{\rho_{water}/\rho_{pol}}{M}\right)^2} \times \left[1 - \frac{M}{M_m \times C_g}\left(-2 \cdot K \cdot a_w \cdot (C_g - 1) - 2 + C_g\right)\right] \tag{13}$$

2.7.3. Calculation of water vapor diffusion coefficient

The water vapor diffusion coefficient (D) was determined at each a_w value from water vapor kinetic measurements. Water vapor sorption data can be converted to a non-dimensional form through Equation (14):

$$\frac{m_t}{m_\infty} = \left(\frac{m_{uptake}(t)}{m_{final} - m_{initial}}\right) \tag{14}$$

where m_t is the water uptake at time t, m_∞ at equilibrium, and $m_{initial}$, m_{final}, m_{uptake}, the mass at the beginning and at the end of each sorption step, and at time t.

By making the hypothesis that the diffusion is isotropic, and that the film is a plan sheet of material of thickness L, Crank equation defines diffusion by Equation (15) [44], with D, the Fickian diffusion, constant.

$$\frac{m_t}{m_\infty} = 1 - \sum_{n=0}^{\infty} \frac{8}{(2n+1)^2\pi^2} exp\left(\frac{-D(2n+1)^2\pi^2 t}{L^2}\right) \tag{15}$$

For each a_w step, the D value was determined as the average value over the interval. Therefore, D values are plotted at the midpoint of each interval. Identification of D was performed using MATLAB, by solving a nonlinear least-squared fitting problem with the *lsqnonlin* function, based on the Levenberg-Marquardt algorithm.

2.8. Water vapor permeability

Water vapor permeability experiments were conducted on a Permatran W 3/33 machine from Mocon. The device is equipped with an infrared sensor, previously calibrated by using reference polyethylene terephthalate (PET) films. Measurements were performed on self-supported films at 25 °C and 40 °C. The film is placed between two compartments. In the upstream compartment, water vapor is introduced at a given pressure, imposing a specific relative humidity, here 50 %. Downstream compartment is flushed with dry nitrogen, which is the carrier gas, transporting permeating water molecules through the film to the IR detector.

Knowing values of thickness of the sample, and of the steady-state water flux, *WVP* values are determined through Equation (16):

$$WVP = \frac{J_{st} \times e}{\Delta P} \tag{16}$$

where e is the film thickness in cm, ΔP the pressure difference between upstream and downstream compartments in cmHg, J_{st} the steady-state flux in g.m^{-2}.day^{-1}. Permeability coefficients P will be expressed in barrer (1 barrer = $10^{-10}\ \frac{cm^3_{STP} \times cm}{cm^2 \times s \times cm_{Hg}}$).

3. Results and discussion

3.1. CMC functionalization by P-3CR

As previously reported, carboxylated polysaccharides (and derivatives) can be successfully functionalized by Passerini three-component in aqueous medium, and various aldehydes and isocyanides have been conveniently employed, to impart polysaccharides with hydrophobic or thermosensitive features [38,40–42]. Herein, we seek to leverage the twofold hydrophobic character of CMC to generate self-standing films with potentially enhanced water vapor barrier properties. CMC modified by P-3CR by three aliphatic aldehydes of different structure and hydrophobicity: butyraldehyde (But), heptaldehyde (Hept), and 5-norbornene-2-carboxaldehyde (Norb), concomitantly with *tert*-butyl isocyanide (*Tert*B), were used. To this end, previously published optimized reaction experimental conditions were employed, and P-3CR reactions were accordingly conducted on partially acidified CMC (Fig. 1). Four syntheses were undertaken, as gathered in Table 1.

The obtained Passerini-modified compounds were thoroughly characterized by ^{1}H NMR spectroscopy. The spectra are available in Supporting Information (Fig. S1–S3). They confirmed the high purity of the final products, showing no evidence of residual reagents, and demonstrated the successful covalent grafting through the clear identification of signals corresponding to the α-acyloxycarboxamide functional group, as previously detailed in a recent publication [42]. The Passerini-modified compounds were obtained in high mass yields (up to 99 %), highlighting the efficiency of the purification steps. Thus, different batches of CMC derivatives modified by *Tert*B and various aldehydes, with different *DS*, were synthesized. Note that the use of heptaldehyde (Table 1, entry 4) leads to a lower *DS* of 0.17, due to its low water solubility linked to the strong hydrophobicity of the C7 chain. In same time, for this product, a low *DS* was targeted to maintain the water solubility of the modified CMC product, that is required to carry out the solvent casting step. It is important to mention that these dually hydrophobized products were previously spin-coated onto silicon wafers and wettability measurements have evidenced that the Passerini grafting of aliphatic residues allowed to impart films with enhanced hydrophobic character [42].

3.2. Macroscopic behavior of the self-supported CMC-based films in liquid water

Unmodified CMC and Passerini-modified CMC derivatives were processed as self-supported films by solvent casting of aqueous solutions. The resulting films were easily handled manually and their behavior in liquid water was visually observed in order to qualitatively evaluate the effect of the presence of the hydrophobic side substituents on dimensional stability. For this purpose, the films were immersed in water at 25 °C and were optically inspected over time. The pictures are provided in Fig. 2.

As shown in Fig. 2, the unmodified CMC film rapidly dissolved in water, only after few minutes of immersion. In contrast, the dually hydrophobized CMC-But-*Tert*B$_{0.31}$ film initially folded onto itself, which could be explained either by a slight swelling difference between the two sides of the film, or by a minimization of its contact area with water due to its enhanced hydrophobic character. Over time, the dimension of all dually hydrophobized CMC-based films increased, due to the progressive water penetration within the film. This difference of behavior in water naturally stems from the grafting of both aliphatic chain and *tert*-butyl groups, that, as expected, confers an increased hydrophobicity to the films. All of them maintained their integrity within 5 days.

3.3. Nanostructure of the self-supported CMC-based films

Small Angle X-ray Scattering (SAXS) measurements were conducted on CMC and Passerini modified-CMC self-supported films. These analyses were performed at different temperatures and water activities, allowing to assess the effect of environmental conditions on the internal nanostructure of the CMC-based materials. Directly *in situ* in the SAXS chamber, the self-handled films were therefore successively exposed to the following conditions: 21.3 °C/a_w = 0, 21.3 °C/a_w = 0.90, 45.5 °C/a_w = 0.90, and 21.4 °C/a_w = 0.90. The samples were left to equilibrate under the specified environmental conditions for 8 h prior to SAXS measurement. This sequential approach was applied in order to monitor a possible structural evolution upon various water activities and temperatures. Note that the humidity stage did not allow to do analysis at temperature higher than 45.5 °C. The scattering diagrams $I = f(q)$ of CMC, CMC-But-*Tert*B$_{0.31}$, CMC-Hept-*Tert*B$_{0.17}$, CMC-Norb-*Tert*B$_{0.35}$ at 21.3 °C and a_w = 0 are represented in Fig. 3, as typical examples.

At 21.3 °C and a_w = 0, significant differences of the scattering intensity profiles are observed between the different samples (Fig. 3). Notably, all Passerini-modified samples exhibit a distinct structural peak, suggesting a nanoscale organization. The CMC reference does not exhibit such a peak, confirming that it is a direct consequence of the dual Passerini ligation. This structural organization is probably driven by interchain hydrophobic interactions between the grafted side-chains, thus leading to the formation of (nano)aggregated hydrophobic domains. Moreover, the drying step after the casting stage may favor the self-association process. A schematic representation of the possible formation of associated domains is depicted in Fig. 4, and in this scheme, hydrophobic domains are spaced by d that corresponds to the average center-to-center distance between two adjacent aggregated zones (see explanations below).

The experimental scattering data were fitted by using a composite function, chosen to fit multiscale structural features of the films. I(q) was

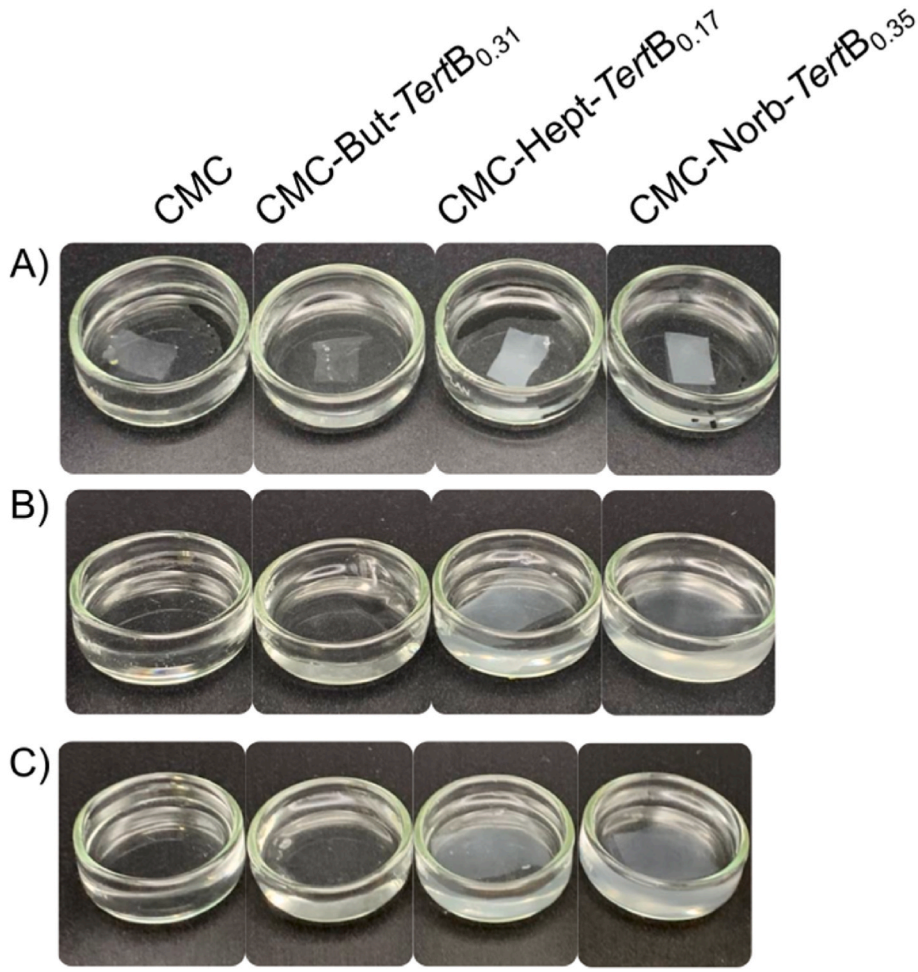

Fig. 2. Pictures of CMC, CMC-But-*Tert*$B_{0.31}$, CMC-Hept-*Tert*$B_{0.17}$ and CMC-Norb-*Tert*$B_{0.35}$ films immerged in water at 25 °C, after A) 1 min, B) 4 h and C) 5 days.

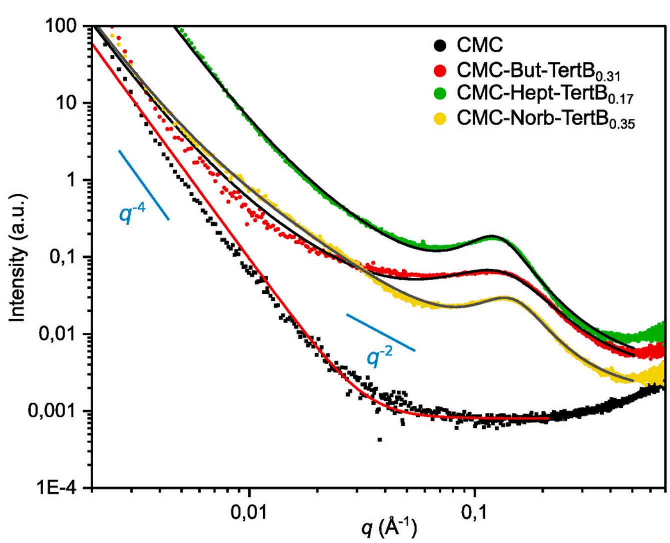

Fig. 3. Evolution of I as a function of q for self-supported films of CMC derivatives of various structure, at 21.3 °C and $a_w = 0$. SAXS curves have been shifted for clarity. Experimental data are shown as scattered points, and modeling data are displayed in solid lines.

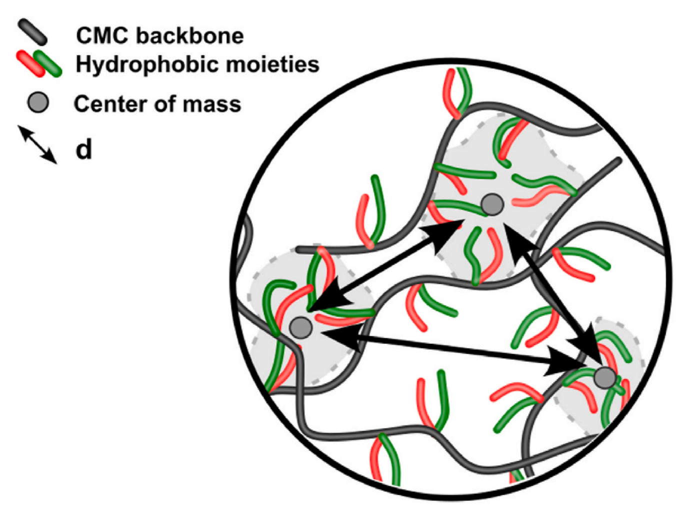

Fig. 4. Schematic representation of hydrophobic domains developed within the Passerini modified CMC-based self-supported films.

modeled as the sum: $I(q) = \frac{A}{q^4} + \frac{B}{q^2} + \frac{C}{1+(2(q-q_0)/w)^2} + I_W$ [45]. The first term in q^{-4} describes the behavior in the low q region ($q < 0.01$ Å^{-1}), where the variation is the steepest, capturing large-scale heterogeneities, such as surface roughness or microcracks present within the films. The second term in q^{-2} observed for the intermediate q region (1×10^{-2} Å$^{-1} < q < 5 \times 10^{-1}$ Å^{-1}) can be associated with fractal-like behavior within the samples. Then, a clearly identifiable structure peak is observed for the Passerini modified CMC samples, which was modeled by using a Lorentzian function, where the parameter $d = 2\pi/q_0$ represents the average center-to-center distance between adjacent hydrophobic domains, while w provides an indication of the degree of order, w being the full width at half maximum of the peak: the lower the w value, the more ordered the system. Finally, I_W corresponds to the intensity at high q values, where the SAXS signal merges into the Wide-Angle X-ray Scattering regime. In this region, the signal appears as a flat and low-intensity baseline, which was simply modeled by using a constant term. All of the parameters of fitted data are available in SI part (Table S1).

To simply analyze the structure peak, $q \times I(q)$ *vs.* q has been plotted (Figs. S4–S6): this representation allows to facilitate the peak characterization and its analysis with a Lorentzian fitting gives access to q_0 and w. These parameters are available in Table 2 for all samples and for the different employed environmental conditions.

The position (q_0) and the width at half maximum of the peak (w) varied slightly as a function of the structure of the grafted substituent. At 21.3 °C and $a_W = 0$, the structure peak for CMC-But-*Tert*B, CMC-Hept-*Tert*B, and CMC-Norb-*Tert*B appear respectively at a q value of $q_0 = 0.146$ Å^{-1}, $q_0 = 0.133$ Å^{-1} and $q_0 = 0.155$ Å^{-1}, corresponding to average center-to-center distances (see Fig. 4) of $d = 4.29$ nm, $d = 4.73$ nm, and $d = 4.06$ nm (Table 2, lines 1, 5 and 9). The higher distance value obtained for CMC-Hept-*Tert*B (4.73 nm) suggests that the grafting of longer aliphatic chains in C7 leads to the formation a little larger and/or the presence of more extended aggregated domains within the film matrix. The slightly narrower peak width of 0.092 Å^{-1} observed for this sample compared to the other derivatives (w(CMC-But-*Tert*B) = 0.133 Å^{-1}, w(CMC-Norb-*Tert*B) = 0.102 Å^{-1}) (Table 2, lines 1, 5 and 9) also indicates a higher degree of structural regularity, with the formation of more uniform interdomain spacing. The Passerini ligation of shorter substituents yields smaller and less ordered aggregated domains. Although variations of SAXS parameters through changes in conditions (of T and RH) remain subtle, some slight differences emerge. Increasing the a_w from 0 to 0.9 (Table 2, lines 2, 6, 10) leads to a narrowing of the scattering peak width for all samples, suggesting a higher degree of organization within the film, possibly due to the molecular mobility that is promoted probably by the increase of water uptake (see *vide supra*). This change is accompanied by a slight increase in the center-to-center d distance for all samples, which may be attributed to a partial swelling phenomenon and/or to the formation of water clusters, both being favored upon higher a_W value. When temperature rises, d continues to increase for all samples (Table 2, lines 3, 7, 11), probably due to a thermo-induced promotion of the local mobility.

SAXS measurements showed that the self-standing films made of Passerini-modified CMC derivatives exhibit a nanoscale structural organization mediated by the formation of interchain hydrophobic interactions leading to aggregated domains; their size and degree of order/regularity depend on the chemical structure of the grafted substituents. Moreover, upon increasing water activity and temperature, a slight rearrangement of the (nano)domains within the films appears, probably due to a partial increase of water uptake, which plays on both overall mobility and the possible formation of water nanoclusters. The internal organization, induced by the presence of lateral aliphatic moieties and its dependence with water uptake, allows us to anticipate that an effect on water vapor transport properties is likely to occur.

Table 2
Lorentzian fitting parameters for structure peaks observed through SAXS of Passerini modified CMC-based self-supported films at various temperatures and a_W values.

Sample	Entry	T (°C)	a_W	q_0 (Å^{-1})	d (nm)	w (Å^{-1})	R^2
CMC-But-*Tert*B$_{0.31}$	1	21.3	0	0.146	4.29	0.133	0.99
	2	21.3	0.9	0.141	4.45	0.127	0.99
	3	45.5	0.9	0.135	4.64	0.120	0.99
	4	21.4	0.9	0.135	4.66	0.122	0.99
CMC-Hept-*Tert*B$_{0.17}$	5	21.3	0	0.133	4.73	0.092	0.99
	6	21.3	0.9	0.131	4.79	0.086	0.99
	7	45.5	0.90	0.129	4.86	0.092	0.99
	8	21.4	0.9	0.129	4.88	0.092	0.99
CMC-Norb-*Tert*B$_{0.35}$	9	21.3	0.9	0.155	4.06	0.102	0.98
	10	21.3	0.9	0.152	4.14	0.072	0.97
	11	45.5	0.9	0.150	4.18	0.090	0.99
	12	21.4	0.9	0.161	3.91	0.091	0.99

3.4. Characterization of the self-supported CMC-based films by dynamic water vapor sorption (DVS) measurements

3.4.1. Water sorption isotherms

A comprehensive study of water vapor sorption is essential to evaluate the impact of the chemical modifications on the equilibrium and dynamic moisture uptake of self-supported films [46,47]. To this end, Dynamic Vapor Sorption (DVS) experiments were conducted on casted films of unmodified and modified CMC-based derivatives at 20 °C from $a_w = 0$ to 0.95. Examples of evolutions of the mass uptake in water vapor of the films as function of the time, at 20 °C of CMC and CMC-But--*Tert*B$_{0.31}$ are given in Fig. S7.

Water vapor sorption isotherms were obtained by calculating the equilibrium moisture content (M) at each water activity level, ranging from $a_w = 0$ to $a_w = 0.95$, for CMC and Passerini-modified CMC films at 20 °C (Fig. 5A). For two samples, namely CMC and CMC-But-*Tert*B$_{0.31}$, the sorption reversibility was examined by performing a complete sorption–desorption cycle (Fig. 5B).

Sorption isotherms of CMC and modified CMC derivatives globally follow a similar evolution but with different levels of moisture content, depending on the type of film (Fig. 5). The curve profile corresponds to a classification of type II of BET (Brunauer, Emmett, Teller) model; his profile is typical of hydrophilic polymers able to develop strong interactions with water, like for example wheat gluten [48], starch [49], flax fibers [50], cellulose whiskers, or cellulose microfibrils [51]. It is characterized by three distinct phases. At low water activity (a_w) (domain I in Fig. 5A), the isotherm is slightly concave. This suggests that water molecules initially bind to specific sorption sites of the polymer chains, *via* strong hydrogen bonds, thus forming a monolayer by following a Langmuir-type adsorption mechanism. However, the concavity of the curve is not very pronounced, which could be explained by a fewer sorption sites. In the intermediate range of a_w (domain II in Fig. 5A), the evolution of moisture content as a function of water activity becomes linear. This linear trend indicates a random sorption, for which water vapor molecules interact not only with the suited sites of the polysaccharide chains but also with the previously adsorbed water sub-layers. This behavior follows Henry's law and reflects the transition from monolayer to multilayer adsorption, characteristic of the BET model. At high a_W (domain III, Fig. 5A), the isotherm becomes convex, with a more pronounced increase in water uptake, for all derivatives. This evolution is generally associated to water clustering. In the case of CMC, at $a_W = 0.95$, equilibrium moisture content rises to a maximum of 1 g g^{-1}, whereas this value does not exceed 0.5 g g^{-1} for grafted samples.

The water vapor sorption is thus reduced for CMC dually functionalized, compared to unmodified CMC and this effect is particularly pronounced at high water activities. This decrease in sorption can be ascribed to *i*) the lowering of the number of –COOH hydrophilic sites available for water binding caused by the Passerini twofold grafting, *ii*) the increase of the hydrophobic character of the film brought by the presence of the two aliphatic substituents, and *iii*) the formation of the internal organization based on the formation of hydrophobic nanodomains within the films, as demonstrated by SAXS analysis. Results can be compared with previously reported data on carboxymethylstarch grafted with methyl methacrylate (MMA) [52], and on cellulose nanocrystals functionalized in surface by aromatic groups [53], which both exhibit a reduced water vapor uptake due to the presence of

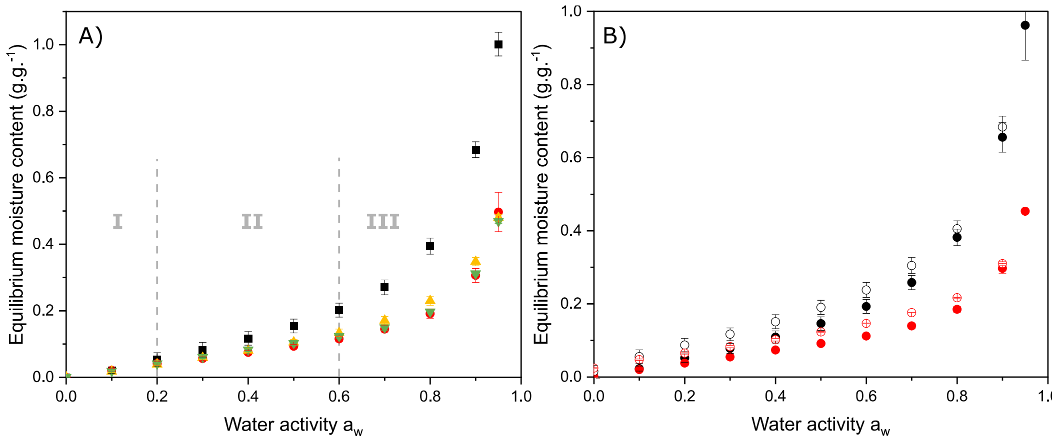

Fig. 5. A) Water vapor sorption isotherms of CMC (■), CMC-But-*Tert*B$_{0.31}$ (●), CMC-Norb-*Tert*B$_{0.35}$ (▲), CMC-Hept-*Tert*B$_{0.17}$ (▼) films measured at 20 °C, B) Water vapor sorption (●, ●) and desorption (○, ○) isotherms of CMC (in black) and CMC-But-*Tert*B$_{0.31}$ (in red) measured at 20 °C. For some experiments, error bars are enclosed within the thickness of the symbols.

hydrophobic residues. At $a_w = 0.70$, the modified- CMC derivatives exhibited water vapor uptake values ranging from 0.14 to 0.17 g g^{-1}, values slightly higher than those reported for MMA-grafted starches (approximately 0.06 g g^{-1}). However, it is worth noting that native carboxymethyl starch displays water vapor uptake lower than native CMC (values of 0.17 g g^{-1} at $a_w = 0.70$ for native starch and 0.27 g g^{-1} for CMC in the same conditions).

Interestingly, despite its lower degree of substitution ($DS = 0.17$ against $DS \approx 0.3$), the CMC-Hept-*Tert*B film appears to perform as well as – or even better than – other modified polysaccharides in terms of water vapor uptake. This performance can be reasonably attributed to the presence of longer aliphatic C7 chains that efficiently interferes with water sorption, by limiting the accessibility of water molecules to polar and hydrophilic functional groups (-OH, and especially –COOH or –COO$^-$ groups).

Fig. 5B shows the presence of a slight hysteresis between the sorption and desorption cycle for CMC and CMC-But-*Tert*B$_{0.31}$, that can be due to a conformational change associated with the swelling capacity of hydrated CMC backbone. Water molecules act as a kind of plasticizing agents which disrupt the intermolecular hydrogen bonds upon water uptake, thus leading to a partial and irreversible relaxation of polymer chains.

To further understand the water sorption mechanism, the GAB model was applied to experimental isotherm data. This model considers multilayer adsorption phenomena, making it suitable for materials that interact strongly with water such as polysaccharides [54]. Typical examples of the GAB model applied to the sorption isotherm of unmodified and modified CMC films are presented in Fig. 6A) and 6B), respectively, which shows the model's ability to accurately fit the experimental data over the whole range of a_w values. The fitting results for the other samples are available in SI (Figs. S8 and S9). The GAB parameters are namely the monolayer moisture content (M_m), corresponding to the water content required to saturate all primary sorption sites, the corrective factor (k), relative to the adsorption energies of second and subsequent layers, and the Guggenheim constant (C_g), related to the energy associated with the interactions between the penetrant molecules and the first sorption sites.

The GAB parameters were determined by fitting water vapor sorption isotherms of unmodified CMC, CMC-But-*Tert*B$_{0.31}$, CMC-Hept-*Tert*B$_{0.17}$, CMC-Norb-*Tert*B$_{0.35}$ and are summarized in Table 3.

The quality of the fit is confirmed by the mean *MRD* values, which are 11 % or lower, indicating a good agreement between the model and the raw data. The values of the GAB parameters obtained for unmodified CMC are of the same order of magnitude than those reported in the

Table 3
Sorption parameters M_m, k, C_g of CMC-based films determined from the GAB sorption model, at 20 °C.

Entry	Sample	M_m (g.g^{-1})	k	C_g	MRD (%)
1	CMC	1.1×10^{-1}	0.94	5.6	11.6
2	CMC-But-*Tert*B$_{0.31}$	5.2×10^{-2}	0.95	11.3	9.0
3	CMC-Hept-*Tert*B$_{0.17}$	5.0×10^{-2}	0.94	11.0	9.0
4	CMC-Norb-*Tert*B$_{0.35}$	7.7×10^{-2}	0.94	4.4	10.4

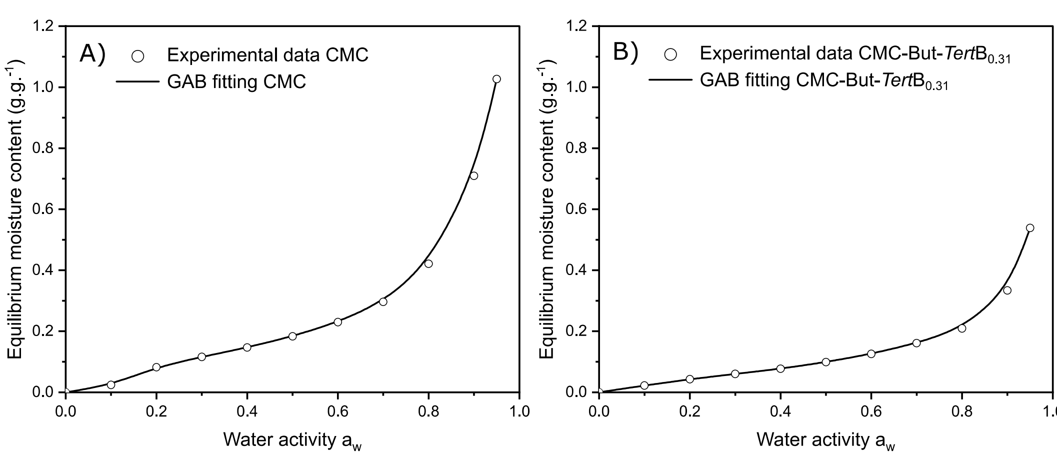

Fig. 6. Experimental data of sorption isotherms (○) and GAB fit (in line), of A) unmodified CMC and B) CMC-But-*Tert*B$_{0.31}$ films at 20 °C.

literature for this polysaccharide [55]. In particular, the water content required to saturate all primary sorption sites, M_m, is found to be 1.1 × 10^{-1} g g^{-1}, which is slightly higher than the value of 9.0 × 10^{-2} g g^{-1}, reported by Torres et al. This value highlights the strong hygroscopic character of CMC, in line with its high degree of carboxymethylation and the abundance of hydrophilic functional groups (-COOH, –COO^-, –OH) capable of interacting with water vapor. The Guggenheim constant C_g, which describes the affinity of water molecules sorbed onto the monolayer, equal to 5.6 for CMC is also close to the value of 5.9 found by Torres et al. The hydrophobic functionalization yields a decrease in M_m for CMC-But-*Tert*$B_{0.31}$ and for CMC-Hept-*Tert*$B_{0.17}$, from 1.1 × 10^{-1} g g^{-1} down to 5.2 × 10^{-2} and 5.0 × 10^{-2} g g^{-1} and an increase in C_g from 5.6 up to approximately 11. Such changes are not so marked for CMC-Norb-*Tert*$B_{0.35}$, a decrease of M_m to 7.7 × 10^{-2} g g^{-1} is observed and no clear effect on C_g can be detected. The k parameter remains approximately constant across all samples ($k \approx 0.95$). A decrease of M_m combined with an increase in C_g is consistent with the consumption of hydrophilic –COOH groups caused by the Passerini grafting, resulting in a reduced number of specific binding sites and an increase in the hydrophobic character of the modified CMC films. A similar distinction between single-layer and multi-layer absorption has also been reported for hemicellulose/PVA composites, where the D'Arcy and Watt model successfully described the coexistence of monolayer adsorption sites and multilayer water clustering [56]. This comparison further supports the interpretation proposed here for CMC-based systems.

Zimm-Lundberg theory was exploited to interpret clustering phenomena within polysaccharide films [50]. This model enables the determination of the mean cluster size (*MCS*) directly from experimental sorption data. In addition, the number of water molecules per sugar unit (*Ni*) can be obtained from the sorption isotherms. The *Ni/MCS* ratio is then calculated to estimate the number of sorption sites per sugar unit. These parameters, *i.e.* *Ni*, *MCS* and *Ni/MCS*, were determined for each water activity, and their evolution are represented in Figure, for CMC, CMC-Hept-*Tert*$B_{0.17}$, CMC-Norb-*Tert*$B_{0.35}$, and CMC-But-*Tert*$B_{0.31}$, at 20 °C.

By definition, *Ni* values follow the same trends as the moisture content, and thus a decrease of *Ni* values is observed for all grafted samples in Fig. 7A. For each sample, at low a_w ($a_w < 0.5$), *MCS* is close to 1 (Fig. 7B), and *Ni/MCS* ratio increases with the water activity (Fig. 7C), suggesting that water molecules are primary sorbed individually onto the specific sites, in accordance with the Henry-type sorption behavior. At intermediate a_w ($0.5 > a_w < 0.75$), *MCS* increases (Fig. 7B) and *Ni/MCS* reaches a plateau (Fig. 7C), reflecting the saturation of available sorption sites and meaning that the sorption of additional water molecules predominantly form clusters [50]. For unmodified CMC, the plateau reached by *Ni/MCS* indicates approximately 3 sorption sites per sugar unit, which is consistent with the 3 available functional groups (-OH and –COOH) per one sugar unit, able and available to develop hydrogen bonds with water molecules. As expected, the covalent introduction of hydrophobic substituents leads to a significant reduction in the *Ni/MCS* value at the plateau, reflecting a decrease in the number of available sorption sites to around 1.5 per sugar unit. At higher values of a_w ($a_w > 0.75$), deviations from the plateau reflect a deviation from the Zimm-Lundberg theory, which may be attributed to a too strongly hydrophilic character of the sample. This deviation is more pronounced in the case of unmodified CMC (Fig. 7c), probably due to its greater sensitivity to moisture. *MCS* values indicate the formation of clusters of up to 5 water molecules in the case of CMC-But-*Tert*$B_{0.31}$ and CMC-Hept-*Tert*$B_{0.17}$, value larger than those obtained for unmodified CMC, with *MCS* = 3 at a_w = 0.95. The hydrophobic functionalization diminishes the number of available sorption sites and enhances hydrophobicity, in which case water molecules tend to form larger aggregates

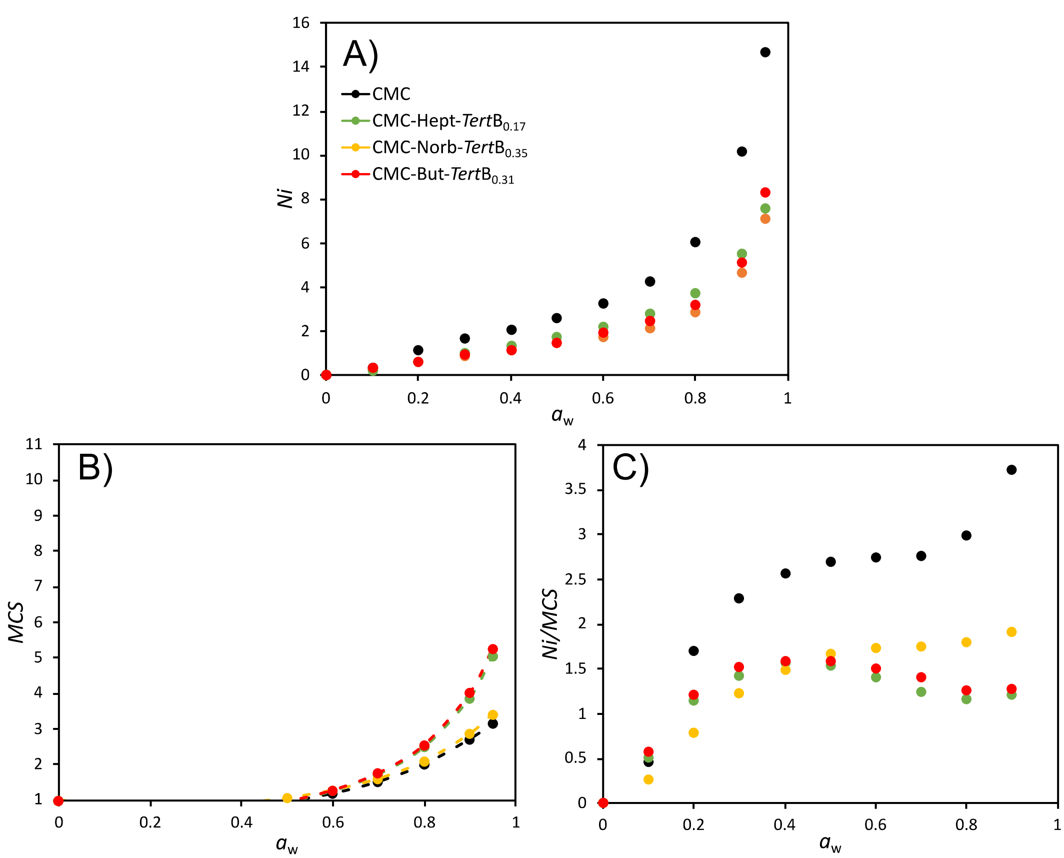

Fig. 7. A) Number of water molecules per sugar unit (*Ni*), B) mean cluster size (*MCS*), and C) number of sorption sites bound to water molecules per sugar unit (*Ni/MCS*) as a function of a_w.

mediated by numerous water-water interactions.

3.4.2. Determination of water vapor diffusion coefficient

By considering that the transport mechanism follows a Fickian behavior, effective water vapor diffusion coefficient (*D)* values were determined for CMC, CMC-But-*Tert*$B_{0.31}$, CMC-But-*Tert*$B_{0.35}$, CMC-Hept-*Tert*$B_{0.17}$, and are plotted as a function of the water activity (Fig. 8)

For all samples, the water activity-dependence of *D* follows a bell-shaped trend, by reaching a maximum at $a_w = 0.6$. This two-stage type behavior is consistent with results reported in literature, notably for natural fiber-based materials by Thoury-Monbrun et al. (cellulose) [44], Wolf et al. (wheat straw) [57], and Gouanvé et al. (flax fiber) [50]. The initial increase in *D* is attributed to a slight plasticization of the polysaccharide chains: the water sorption leads to a partial disruption of intermolecular interactions, which enhances the overall mobility and thus facilitates and favors water diffusion and transport. In contrast, the decrease in *D* at high a_w (from 0.6) is commonly linked to the formation of water clusters. In this regime, water–water interactions dominate, leading to reduced (macro)molecular mobility. This interpretation further supports GAB model observations through high *MCS* values, and the S shape of the isotherm previously described.

For $a_w < 0.5$, the Passerini-hydrophobized CMC-based films exhibit slightly lower *D*, compared to unmodified CMC, that could mean that the dual hydrophobization limits the plasticizing effect. For $a_w > 0.5$, *D* values obtained for these samples are rather similar.

DVS measurements provided valuable insights into water molecules transport inside Passerini-modified CMC films. The chemical ligation significantly reduces the equilibrium moisture content at high water activity ($a_w = 0.95$), from 1.0 g g^{-1} for native CMC to less than 0.50 g g^{-1} for all grafted samples. The application of the GAB model reveals this reduction, due to the consumption of –COOH groups, and to the introduction of hydrophobic moieties (e.g., -But-*Tert*B, -Hept-*Tert*B, -Norb-*Tert*B). Hydrophobic domains are formed, yielding a decrease of accessible sorption sites, from approximately 3 to 1.5 sites per sugar unit and an increase in the size of water clusters with an *MCS* value evolution of approximately 3–5 water molecules at $a_w = 0.95$.

3.5. Water vapor permeability measurements

To fully assess the films barrier performances, water vapor permeability (WVP) measurements are essential, and allow to investigate how the nanostructural features evidenced by SAXS and sorption phenomena observed by DVS can be translated to dynamic measurements.

The WVP of each film was measured for three water activities (a_w = 0.5, 0.6 and 0.7), at two temperatures (25 °C and 40 °C), and the results are gathered in Table 4.

For unmodified CMC, very high WVP values are found that stem from its highly hydrophilic and hygroscopic character. At 25 °C, the WVP of CMC film substantially increases with a_w, reaching 74 800 barrer at a_w

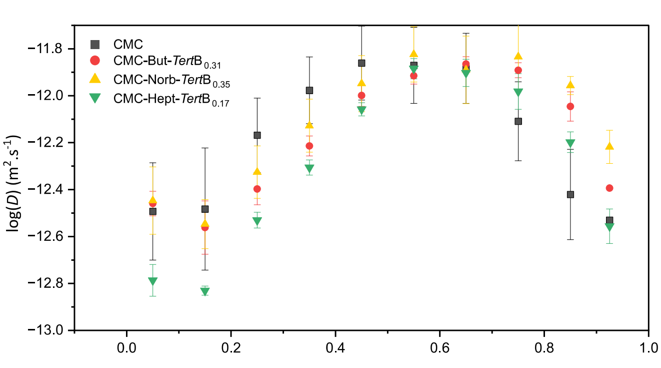

Fig. 8. Water vapor diffusion of CMC (■), CMC-But-*Tert*$B_{0.31}$ (●), CMC-Norb-*Tert*$B_{0.35}$ (▲), CMC-Hept-*Tert*$B_{0.17}$ (▼) at 20 °C.

Table 4
Water vapor permeability of CMC-based self-supported films as a function of water activity, a 10 % uncertainty can be considered.

	T (°C)	WVP (barrer)		
		a_w		
		0.5	0.6	0.7
CMC	25	44 200	52 100	74 800
	40	34 000	N/A	N/A
CMC-But-*Tert*$B_{0.13}$	25	2500	3600	6100
	40	2800	4900	10 900
CMC-But-*Tert*$B_{0.30}$	25	14 300	20 100	36 300
	40	8500	11 500	25 900
CMC-Norb-*Tert*$B_{0.35}$	25	6800	8200	22 500
	40	7500	10 200	22 200

N/A: non-applicable, due to too high gas fluxes which exceed the detection range.

= 0.7. This increase is a general trend observed for polysaccharide-based materials and was previously reported in studies on starch systems [58, 59]. The obtained values are close to the ones found in literature, for example at $a_w = 0.5$, the CMC films exhibit WVP = 52 100 barrer, while Li et al. found WVP = 59 700 for CMC films of *DS* = 0.9 in the same conditions of a_w [60]. As shown in Table 4, the Passerini functionalization considerably decreases the WVP values. For instance, CMC-But-*Tert*$B_{0,13}$ shows a WVP of 6100 barrer at 25 °C, at $a_w = 0.7$, compared to 74 800 for unmodified CMC. CMC-Norb-*Tert*$B_{0.35}$ and CMC-But-*Tert*$B_{0.30}$ -based films exhibit higher WVP values (respectively 36 300 and 22 500barrer, for $a_w = 0.5$) than CMC-But-*Tert*$B_{0.13}$. This could originate from a difference in the internal organization of the films. From a general statement, the permeability, *P*, is governed by $P = S \times D$, where *S* is the permeate solubility and *D* its diffusion coefficient. Permeability is thus related to the chemical affinities of water molecules for the film and to the (macro)molecular mobility of the components of the film. The Passerini modification unambiguously plays on the *S* parameter: the dual chemical grafting leads not only to a decrease of the number of sites of sorption (decrease of *Ni/MCS*, as shown in Fig. 7C)) due to the substitution of –COOH groups by aliphatic substituents but also to a significative enhancement of the hydrophobic character of the derivatives, as previously reported by the increase of water contact angles measured on deposited films [42]. This effect is reflected by the decreasing in equilibrium moisture content, as observed by the DVS experiments represented in Fig. 5. Moreover, even if measurements of effective diffusion coefficients of water vapor are not unequivocally conclusive (Fig. 8), SAXS analysis have demonstrated that the ligation of the two aliphatic moieties yields a peculiar nano-organization within the film, linked to the formation of aggregated nanodomains stabilized by hydrophobic interactions. Thus, we can do the assumption that such internal organization partially restricts the overall mobility, which may decrease the transport of water vapor. Furthermore, the presence of water clusters, observed for chemically modified CMC derivatives at high a_w values may contribute to the decrease of the WVP. To sum up, the decreasing in water vapor permeability seems to be predominantly dictated by the sorption phenomenon, and by the mobility restriction to a letter extent. When compared with WVP values reported in the literature, the Passerini-modified CMC derivatives, and especially CMC-But-*Tert*$B_{0.13}$, exhibit unprecedently low WVP for polysaccharide films. For instance, the incorporation of nanoclay fillers, as studied by Fiori et al., reduced the WVP of CMC films from 35 300 barrer down to 7600 barrer. Similarly, Li et al. reported a reduction from 15 300 to approximately 8300 barrer, by incorporating CNCs derived from pea hulls (20 °C, $a_w = 0.75$), while El Miri et al. stated a decrease from 62 130 to 18 400 barrer for the inclusion of CNCs (32 °C, $a_w = 0.5$) into CMC.

Additionally, at high $a_w = 0.7$, the increase in temperature at 40 °C globally leads to an increase in WVP for each sample. Generally, the increase of WVP values is due to the increase of the overall mobility that promotes and accelerates the water permeation within the films. WVP values could not be measured at 40 °C, at $a_w = 0.6$ and 0.7, for unmodified CMC. For this sample, too high gas fluxes are required, which is not possible to do with this device. This indirectly reveals that very high WVP value should be obtained. Therefore, this shows that the Passerini coupling allows us for maintaining relatively low water vapor permeability at higher T°, compared to neat polysaccharide, which is a promising prerequisite for storage of packaging in tropical geographic zones.

4. Conclusion

The P-3CR was shown to be efficient to synthesis dually hydrophobized by aliphatic reactants. The functionalized CMC products were successfully processed into macroscopic homogeneous self-supported films, that can be easily handled. The covalent introduction of aliphatic substituents along the CMC backbone allows to reinforce the cohesion of the films and to impart them with promoted dimensional stability in water liquid. SAXS experiments carried out on the films in different conditions of temperature and a_w unequivocally revealed the formation of a peculiar structure at the nanoscale within the Passerini modified CMC-based films, that is not visible for unmodified CMC. The structure formed results from the development of hydrophobic interchain interactions between the grafted substituents in water, and this aggregation is maintained within the self-supported films. The nature of the grafted side-chains was found to have an influence on the nanostructure. For example, dually hydrophobized derivatives bearing alkyl chains in C4 or cyclic groups yielded smaller/more compact nanodomains than longer alkyl chains in C7, with higher correlation distances. DVS measurements revealed that the Passerini-modified CMC films showed reduced water uptake compared to unmodified CMC, especially at high relative humidity. The GAB model showed a reduction in monolayer capacity and a Zimm-Lundberg analysis confirmed a decrease in the number of accessible sites of sorption, consistent with the consumption of –COOH groups during the chemical reaction and their replacement by sterically hindering substituents. Water clustering occurred for all samples above $a_w = 0.6$, with an increase in cluster size upon grafting, reaching up 5 molecules per cluster in the case of modified-CMC films, likely resulting from the diminution in the number of available sorption sites, and the global enhancement of hydrophobicity, which promotes water-water interactions. Overall, the Passerini functionalization drastically improved the water vapor permeability of CMC films. While CMC showed a high WVP, it was substantially reduced by the grafting of butyl and *tert*-butyl groups and the CMC-But-*Tert*$B_{0.13}$ derivative seems to be the most relevant product among the tested ones. This effect is mainly dictated by the decrease of the number of sites of sorption and by the hydrophobization, with probably an effect of the internal structuration, to a lesser extent. Moreover, low WVP values are maintained at higher temperature, which is a key result when a storage in tropical conditions is desired. It would be interesting to dually modify CMC with even more hydrophobic reagents, such as C7, C8, or C11 aliphatic chains, but the challenge lies in maintaining the solubility/dispersibility of the modified products in water. These results position the Passerini-modified CMC films as promising candidates for designing biobased barrier materials. Their tunable dual functionalization, offered by the Passerini three-component reaction, unlocks new opportunities for the use of CMC derivatives in sustainable packaging applications, used as self-supported films or as functional coatings for paper and board packaging.

CRediT authorship contribution statement

Clémence Vuillet: Writing – original draft, Methodology, Investigation, Data curation, Conceptualization. **Valérie Guillard:** Writing – review & editing, Validation, Investigation. **Hélène Angellier-Coussy:** Writing – review & editing, Validation, Investigation. **Guillaume Sudre:** Writing – review & editing, Investigation. **Fabrice Gouanvé:** Writing – review & editing, Validation, Methodology, Investigation. **Etienne Fleury:** Writing – review & editing, Supervision, Investigation. **Aurélia Charlot:** Writing – review & editing, Validation, Supervision, Project administration, Methodology, Investigation, Funding acquisition, Conceptualization.

Funding sources

This work was supported by the Agence Nationale de la Recherche (ANR) in France as part of the project MODERATO (Grant number: ANR-21-CE06-0035).

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: AURELIA CHARLOT reports financial support was provided by ANR. CHARLOT AURELIA reports a relationship with French National Research Agency that includes: funding grants. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

The authors thank the Agence Nationale de la Recherche (ANR) in France for the financial support of this project MODERATO (Grant number: ANR-21-CE06-0035). The authors acknowledge UAR2045/CSpec-LaDoua for providing access to the NMR facilities and assistance.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.memsci.2025.124851.

Data availability

Data will be made available on request.

References

[1] E. Díaz-Montes, Polysaccharide-based biodegradable films: an alternative in food packaging, Polysaccharides 3 (4) (Dec. 2022), https://doi.org/10.3390/polysaccharides3040044. Art. no. 4.

[2] E. Díaz-Montes, Polysaccharides: sources, characteristics, properties, and their application in biodegradable films, Polysaccharides 3 (3) (Sep. 2022), https://doi.org/10.3390/polysaccharides3030029. Art. no. 3.

[3] R. Thakur, P. Pristijono, C.J. Scarlett, M. Bowyer, S.P. Singh, Q.V. Vuong, Starch-based films: major factors affecting their properties, Int. J. Biol. Macromol. 132 (Jul. 2019) 1079–1089, https://doi.org/10.1016/j.ijbiomac.2019.03.190.

[4] I.P. Butler, R.A. Banta, A.A. Tyuftin, J. Holmes, S. Pathania, J. Kerry, Pectin as a biopolymer source for packaging films using a circular economy approach: origins, extraction, structure and films properties, Food Packag. Shelf Life 40 (Dec. 2023) 101224, https://doi.org/10.1016/j.fpsl.2023.101224.

[5] M. Flórez, E. Guerra-Rodríguez, P. Cazón, M. Vázquez, Chitosan for food packaging: recent advances in active and intelligent films, Food Hydrocoll. 124 (Mar. 2022) 107328, https://doi.org/10.1016/j.foodhyd.2021.107328.

[6] T.S. Parreidt, K. Müller, M. Schmid, Alginate-based edible films and coatings for food packaging applications, Foods 7 (10) (Oct. 2018) 170, https://doi.org/10.3390/foods7100170.

[7] S. Sabu Mathew, A.K. Jaiswal, S. Jaiswal, Carrageenan-based sustainable biomaterials for intelligent food packaging: a review, Carbohydr. Polym. 342 (Oct. 2024) 122267, https://doi.org/10.1016/j.carbpol.2024.122267.

[8] K. Agarwal, Rubeka, S. Jain, M.D. Heirangkhongjam, A. Agarwal, Gellan gum-based films, in: T. Amin, H.R. Naik, S.Z. Hussain, S.M. Wani (Eds.), Polysaccharide Based Films for Food Packaging: Fundamentals, Properties and Applications, Springer Nature, Singapore, 2024, pp. 197–217, https://doi.org/10.1007/978-981-99-4898-7_8.

[9] U. Edlund, Y.Z. Ryberg, A.-C. Albertsson, Barrier films from renewable forestry waste, Biomacromolecules 11 (9) (Sep. 2010) 2532–2538, https://doi.org/10.1021/bm100767g.

[10] M. Anker, J. Berntsen, A.-M. Hermansson, M. Stading, Improved water vapor barrier of whey protein films by addition of an acetylated monoglyceride, Innov. Food Sci. Emerg. Technol. 3 (1) (Apr. 2002) 81–92, https://doi.org/10.1016/S1466-8564(01)00051-0.
[11] M. Petersson, M. Stading, Water vapour permeability and mechanical properties of mixed starch-monoglyceride films and effect of film forming conditions, Food Hydrocoll. 19 (1) (Jan. 2005) 123–132, https://doi.org/10.1016/j.foodhyd.2004.04.021.
[12] L. Saidi, Y. Wang, P.R. Wich, C. Selomulya, Polysaccharide-based edible films — strategies to minimize water vapor permeability, Curr. Opin. Food Sci. 61 (Feb. 2025) 101258, https://doi.org/10.1016/j.cofs.2024.101258.
[13] J. Long, W. Zhang, M. Zhao, C.-Q. Ruan, The reduce of water vapor permeability of polysaccharide-based films in food packaging: a comprehensive review, Carbohydr. Polym. 321 (Dec. 2023) 121267, https://doi.org/10.1016/j.carbpol.2023.121267.
[14] J. Gamelas, E. Ferraz, Composite films based on nanocellulose and nanoclay minerals as high strength materials with gas barrier capabilities: key points and challenges, Bioresources 10 (Aug) (2015), https://doi.org/10.15376/biores.10.4.6310-6313.
[15] M. Quilaqueo Gutiérrez, I. Echeverría, M. Ihl, V. Bifani, A.N. Mauri, Carboxymethylcellulose–montmorillonite nanocomposite films activated with murta (*Ugni molinae* Turcz) leaves extract, Carbohydr. Polym. 87 (2) (Jan. 2012) 1495–1502, https://doi.org/10.1016/j.carbpol.2011.09.040.
[16] A.P.S. de Melo Fiori, P.H. Camani, D. dos Santos Rosa, D.J. Carastan, Combined effects of clay minerals and polyethylene glycol in the mechanical and water barrier properties of carboxymethylcellulose films, Ind. Crop. Prod. 140 (Nov. 2019) 111644, https://doi.org/10.1016/j.indcrop.2019.111644.
[17] S.-H. Shin, S.-J. Kim, S.-H. Lee, K.-M. Park, J. Han, Apple peel and carboxymethylcellulose-based nanocomposite films containing different nanoclays, J. Food Sci. 79 (3) (2014) E342–E353, https://doi.org/10.1111/1750-3841.12356.
[18] H. Li, H. Shi, Y. He, X. Fei, L. Peng, Preparation and characterization of carboxymethyl cellulose-based composite films reinforced by cellulose nanocrystals derived from pea hull waste for food packaging applications, Int. J. Biol. Macromol. 164 (Dec. 2020) 4104–4112, https://doi.org/10.1016/j.ijbiomac.2020.09.010.
[19] A.A. Oun, J.-W. Rhim, Preparation and characterization of sodium carboxymethyl cellulose/cotton linter cellulose nanofibril composite films, Carbohydr. Polym. 127 (Aug. 2015) 101–109, https://doi.org/10.1016/j.carbpol.2015.03.073.
[20] N. El Miri, et al., Bio-nanocomposite films reinforced with cellulose nanocrystals: rheology of film-forming solutions, transparency, water vapor barrier and tensile properties of films, Carbohydr. Polym. 129 (Sep. 2015) 156–167, https://doi.org/10.1016/j.carbpol.2015.04.051.
[21] A.A. Oun, J.-W. Rhim, Isolation of cellulose nanocrystals from grain straws and their use for the preparation of carboxymethyl cellulose-based nanocomposite films, Carbohydr. Polym. 150 (Oct. 2016) 187–200, https://doi.org/10.1016/j.carbpol.2016.05.020.
[22] B. Ghanbarzadeh, H. Almasi, Physical properties of edible emulsified films based on carboxymethyl cellulose and oleic acid, Int. J. Biol. Macromol. 48 (1) (Jan. 2011) 44–49, https://doi.org/10.1016/j.ijbiomac.2010.09.014.
[23] X. Guo, B. Chen, X. Wu, J. Li, Q. Sun, Utilization of cinnamaldehyde and zinc oxide nanoparticles in a carboxymethylcellulose-based composite coating to improve the postharvest quality of cherry tomatoes, Int. J. Biol. Macromol. 160 (Oct. 2020) 175–182, https://doi.org/10.1016/j.ijbiomac.2020.05.201.
[24] K.M. Tavares, A. de Campos, B.R. Luchesi, A.A. Resende, J.E. de Oliveira, J. M. Marconcini, Effect of carboxymethyl cellulose concentration on mechanical and water vapor barrier properties of corn starch films, Carbohydr. Polym. 246 (Oct. 2020) 116521, https://doi.org/10.1016/j.carbpol.2020.116521.
[25] Q. Tong, Q. Xiao, L.-T. Lim, Preparation and properties of pullulan–alginate–carboxymethylcellulose blend films, Food Res. Int. 41 (10) (Dec. 2008) 1007–1014, https://doi.org/10.1016/j.foodres.2008.08.005.
[26] M. Shahbazi, S.J. Ahmadi, A. Seif, G. Rajabzadeh, Carboxymethyl cellulose film modification through surface photo-crosslinking and chemical crosslinking for food packaging applications, Food Hydrocoll. 61 (Dec. 2016) 378–389, https://doi.org/10.1016/j.foodhyd.2016.04.021.
[27] S.R. Kanatt, S.H. Makwana, Development of active, water-resistant carboxymethyl cellulose-poly vinyl Alcohol-*Aloe vera* packaging film, Carbohydr. Polym. 227 (Jan. 2020) 115303, https://doi.org/10.1016/j.carbpol.2019.115303.
[28] M.A.P. Morais, M. Silva, M. Barros, P. Halley, Y. Almeida, G. Vinhas, Impact of citric acid on guar gum carboxymethylcellulose crosslinked blend films, J. Appl. Polym. Sci. 141 (44) (2024) e56162, https://doi.org/10.1002/app.56162.
[29] H. Zhang, Z. Fan, D. Peng, C. Huang, X. Wu, F. Sun, Tunning the hydrophobic performance and thermal stability of pectin film by acetic anhydride esterification, Int. J. Biol. Macromol. 276 (Sep. 2024) 133746, https://doi.org/10.1016/j.ijbiomac.2024.133746.
[30] W. Liu, J. Gu, C. Huang, C. Lai, Z. Ling, Q. Yong, Fabrication of hydrophobic and high-strength packaging films based on the esterification modification of galactomannan, Int. J. Biol. Macromol. 167 (Jan. 2021) 1221–1229, https://doi.org/10.1016/j.ijbiomac.2020.11.076.
[31] M. Shaikh, S. Haider, T.M. Ali, A. Hasnain, Physical, thermal, mechanical and barrier properties of pearl millet starch films as affected by levels of acetylation and hydroxypropylation, Int. J. Biol. Macromol. 124 (Mar. 2019) 209–219, https://doi.org/10.1016/j.ijbiomac.2018.11.135.
[32] A.E.J. De Nooy, G. Masci, V. Crescenzi, Versatile synthesis of polysaccharide hydrogels using the passerini and ugi multicomponent condensations, Macromolecules 32 (4) (Feb. 1999) 1318–1320, https://doi.org/10.1021/ma9815455.
[33] Z. Söyler, K.N. Onwukamike, S. Grelier, E. Grau, H. Cramail, M.A.R. Meier, Sustainable succinylation of cellulose in a CO 2 -based switchable solvent and subsequent Passerini 3-CR and Ugi 4-CR modification, Green Chem. 20 (1) (2018) 214–224, https://doi.org/10.1039/C7GC02577G.
[34] E. Esen, M.A.R. Meier, Sustainable functionalization of 2,3-Dialdehyde cellulose via the passerini three-component reaction, ACS Sustain. Chem. Eng. (Oct. 2020), https://doi.org/10.1021/acssuschemeng.0c06153.
[35] Y.Y. Khine, S. Ganda, M.H. Stenzel, Covalent tethering of temperature responsive pNIPAm onto TEMPO-oxidized cellulose nanofibrils via three-component Passerini reaction, ACS Macro Lett. 7 (4) (Apr. 2018) 412–418, https://doi.org/10.1021/acsmacrolett.8b00051.
[36] Y. Li, M. Milewska, Y.Y. Khine, N. Ariotti, M.H. Stenzel, Trehalose coated nanocellulose to inhibit the infections by S. aureus, Polym. Chem. 13 (11) (Mar. 2022) 1502–1509, https://doi.org/10.1039/D1PY01422F.
[37] O. Hamzah, T. Vandenbrouck, L. Heux, B. Jean, Insight into the hydrophobic functionalization of cellulose microfibrils using the Passerini three-component reaction, Carbohydr. Polym. 341 (Oct. 2024) 122323, https://doi.org/10.1016/j.carbpol.2024.122323.
[38] A. Pettignano, A. Daunay, C. Moreau, B. Cathala, A. Charlot, E. Fleury, Sustainable modification of carboxymethyl cellulose by passerini three-component reaction and subsequent adsorption onto cellulosic substrates, ACS Sustainable Chem. Eng. 7 (17) (Sep. 2019) 14685–14696, https://doi.org/10.1021/acssuschemeng.9b02634.
[39] A. Pettignano, J. Leguy, L. Heux, B. Jean, A. Charlot, E. Fleury, Multifunctionalization of cellulose microfibrils through a cascade pathway entailing the sustainable Passerini multi-component reaction, Green Chem. 22 (20) (Oct. 2020) 7059–7069, https://doi.org/10.1039/D0GC02532A.
[40] L. Remy, G. Sudre, A. Charlot, E. Fleury, α-Substituted ketones as reagent for Passerini modification of carboxymethyl cellulose: toward dually functionalized derivatives and thermo-sensitive chemical hydrogels, Carbohydr. Polym. 320 (Nov. 2023) 121228, https://doi.org/10.1016/j.carbpol.2023.121228.
[41] L. Remy, et al., Thermosensitive cellulosic biohybrid comb copolymers by the Passerini-3CR: 2-fold functionalization and subsequent thermal response, Biomacromolecules (Jun. 2025), https://doi.org/10.1021/acs.biomac.5c00647.
[42] C. Vuillet, C. Le Mouël, D. Albertini, P. Alcouffe, E. Fleury, A. Charlot, Dual functionalization of carboxymethyl cellulose and alginate *via* Passerini three-component reaction to graft two hydrophobic moieties: toward modular thin films, Carbohydr. Polym. 351 (Mar. 2025) 123066, https://doi.org/10.1016/j.carbpol.2024.123066.
[43] E.A. Guggenheim, Applications of Statistical Mechanics, Clarendon P., 1966.
[44] V. Thoury-Monbrun, S. Gaucel, N. Gontard, V. Guillard, H. Angellier-Coussy, Adapting gravimetric sorption analyzer to estimate water vapor diffusivity in micrometric size cellulose particles, Cellulose 26 (16) (Nov. 2019) 8575–8587, https://doi.org/10.1007/s10570-019-02661-0.
[45] G. Lalevée, et al., Highly stretchable hydrogels from complex coacervation of natural polyelectrolytes, Soft Matter 13 (37) (Sep. 2017) 6594–6605, https://doi.org/10.1039/C7SM01215B.
[46] K. Wu, et al., Controllable hydrophilicity-hydrophobicity and related properties of konjac glucomannan and ethyl cellulose composite films, Food Hydrocoll. 79 (Jun. 2018) 301–309, https://doi.org/10.1016/j.foodhyd.2017.12.034.
[47] A. Aljawish, L. Muniglia, A. Klouj, J. Jasniewski, J. Scher, S. Desobry, Characterization of films based on enzymatically modified chitosan derivatives with phenol compounds, Food Hydrocoll. 60 (Oct. 2016) 551–558, https://doi.org/10.1016/j.foodhyd.2016.04.032.
[48] V. Guillard, A. Chevillard, E. Gastaldi, N. Gontard, H. Angellier-Coussy, Water transport mechanisms in wheat gluten based (nano)composite materials, Eur. Polym. J. 49 (6) (Jun. 2013) 1337–1346, https://doi.org/10.1016/j.eurpolymj.2013.02.035.
[49] F. Chivrac, H. Angellier-Coussy, V. Guillard, E. Pollet, L. Avérous, How does water diffuse in starch/montmorillonite nano-biocomposite materials? Carbohydr. Polym. 82 (1) (Aug. 2010) 128–135, https://doi.org/10.1016/j.carbpol.2010.04.036.
[50] F. Gouanvé, S. Marais, A. Bessadok, D. Langevin, C. Morvan, M. Métayer, Study of water sorption in modified flax fibers, J. Appl. Polym. Sci. 101 (6) (2006) 4281–4289, https://doi.org/10.1002/app.23661.
[51] S. Belbekhouche, et al., Water sorption behavior and gas barrier properties of cellulose whiskers and microfibrils films, Carbohydr. Polym. 83 (4) (Feb. 2011) 1740–1748, https://doi.org/10.1016/j.carbpol.2010.10.036.
[52] I. Bravo-Osuna, C. Ferrero, M.R. Jiménez-Castellanos, Water sorption-desorption behaviour of methyl methacrylate-starch copolymers: effect of hydrophobic graft and drying method, Eur. J. Pharm. Biopharm. 59 (3) (Apr. 2005) 537–548, https://doi.org/10.1016/j.ejpb.2004.10.003.
[53] E. Espino-Pérez, et al., Cellulose nanocrystal surface functionalization for the controlled sorption of water and organic vapours, Cellulose 23 (5) (Oct. 2016) 2955–2970, https://doi.org/10.1007/s10570-016-0994-y.
[54] S. Despond, E. Espuche, N. Cartier, A. Domard, Barrier properties of paper–chitosan and paper–chitosan–carnauba wax films, J. Appl. Polym. Sci. 98 (2) (2005) 704–710, https://doi.org/10.1002/app.21754.
[55] M.D. Torres, R. Moreira, F. Chenlo, M.J. Vázquez, Water adsorption isotherms of carboxymethyl cellulose, guar, locust bean, tragacanth and xanthan gums, Carbohydr. Polym. 89 (2) (Jun. 2012) 592–598, https://doi.org/10.1016/j.carbpol.2012.03.055.
[56] J. Liu, Z. Li, J. Gao, Q. Li, Synthesis and mechanistic modeling of pva-based crosslinked films from wood alkaline hemicellulose extracts for enhanced water vapor barrier, J. Membr. Sci. 720 (Mar. 2025) 123772, https://doi.org/10.1016/j.memsci.2025.123772.

[57] C. Wolf, V. Guillard, H. Angellier-Coussy, G.G.D. Silva, N. Gontard, Water vapor sorption and diffusion in wheat straw particles and their impact on the mass transfer properties of biocomposites, J. Appl. Polym. Sci. 133 (16) (2016), https://doi.org/10.1002/app.43329.

[58] C. Masclaux, F. Gouanvé, E. Espuche, Experimental and modelling studies of transport in starch nanocomposite films as affected by relative humidity, J. Membr. Sci. 363 (1) (Nov. 2010) 221–231, https://doi.org/10.1016/j.memsci.2010.07.032.

[59] P. Cheviron, F. Gouanvé, E. Espuche, Starch/silver nanocomposite: effect of thermal treatment temperature on the morphology, oxygen and water transport properties, Carbohydr. Polym. 134 (Dec. 2015) 635–645, https://doi.org/10.1016/j.carbpol.2015.07.067.

[60] Q. Li, Z. Sun, K. Sun, X. Wang, H. Chen, Study on the barrier properties of carboxymethyl cellulose/montmorillonite mulch films regulated by magnetic field, Int. J. Biol. Macromol. 308 (May 2025) 142428, https://doi.org/10.1016/j.ijbiomac.2025.142428.